

NMAM INSTITUTE OF TECHNOLOGY, NITTE
Off-Campus Centre of Nitte(Deemed to be University)

II Sem B.Tech. (CBCS) Mid Semester Examinations - I, February 2023

CV1001-1 – ELEMENTS OF CIVIL ENGINEERING

Duration: 1 Hour

Max. Marks: 20

*Note: Answer any **One** full question from **each Unit**.*

Unit – I

- | | Marks | BT* | CO* | PO* |
|---|--------------|------------|------------|------------|
| 1. a) Explain the following scopes of Civil Engineering,
i) Surveying ii) Transportation Engineering | 4 | L*1 | 1 | 1, 2 |
| b) The forces 20 N, 30 N, 40 N, 50 N and 60 N are acting at one of the angular points of a regular hexagon, towards the other five angular points, taken in order as shown in fig. Q.1(b). Find the magnitude and direction of the resultant force. | 6 | L*3 | 1 | 1, 2 |
| 2. a) State and explain principle of transmissibility of a force with neat sketch. State its limitations | 4 | L*2 | 1 | 1, 2 |
| b) Determine the magnitude of unknown force and resultant force in a system of force as shown in fig. Q.2(b) whose resultant is a horizontal force. | 6 | L*3 | 1 | 1, 2 |

Unit – II

- | | | | | |
|---|---|-----|---|------|
| 3. a) Explain equivalent force-couple system with the help of sketches. | 3 | L*2 | 2 | 1, 2 |
| b) Sketch the resultant of force system acting on the plate element with respect to point A shown in fig. Q.3(b). Given the grid size is 1 m x 1 m. | 7 | L*3 | 2 | 1, 2 |
| 4. a) State and prove Varignon's theorem | 4 | L*3 | 2 | 1, 2 |
| b) Determine the weight W1 and W2 such that string BC is horizontal as shown in fig. Q.4(b), Assume the pulley to be frictionless and mass less. | 6 | L*3 | 2 | 1, 2 |

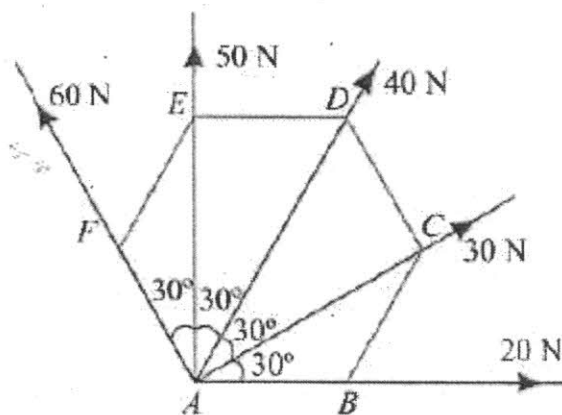


Fig. Q.1(b)

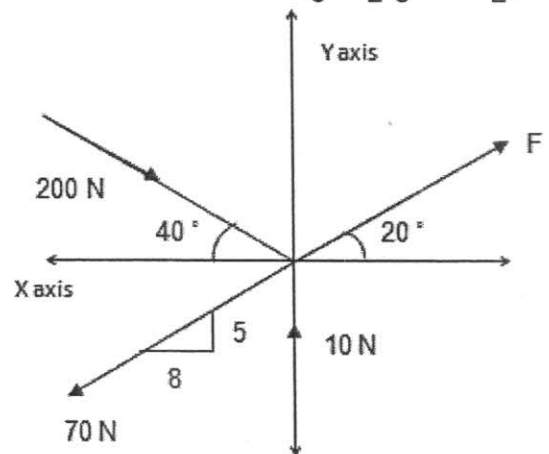


Fig. Q.2(b)

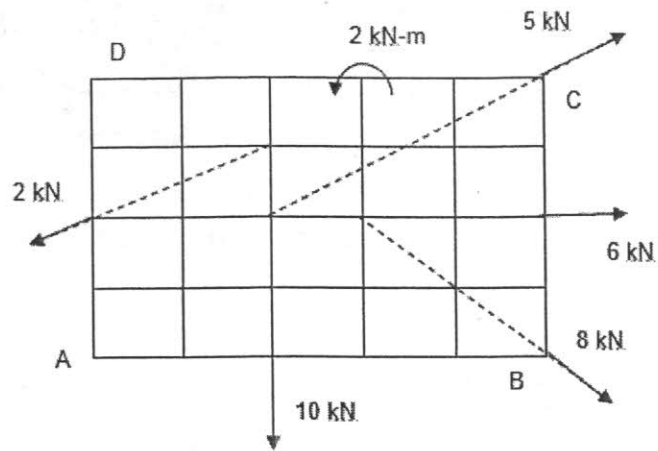


Fig. Q.3(b)

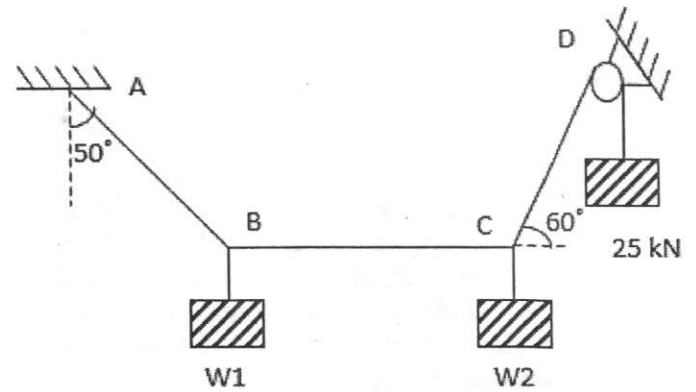


Fig. Q.4(b)

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
